

Tutorial Sheet 1 Solutions
ECE F434: Digital Signal Processing
07/08/2025

1. Answer the following Questions

(a) Check the following functions for linearity.

i. $y[n] = Ax[n] + B$ where A and B are constants.

Answer: Use the linearity property $y[a + b] = y[a] + y[b]$.

$$\text{Let: } y_1[n] = Ax_1[n] + B \text{ and } y_2[n] = Ax_2[n] + B$$

Additivity Test:

$$y[n] = A(x_1[n] + x_2[n]) + B$$

$$y[n] = Ax_1[n] + B + Ax_2[n] + B = A(x_1[n] + x_2[n]) + 2B$$

You'll see that the two equations are not equal, as such this function is non linear.

ii. $y[n] = x[n^2]$

Answer: Similarly to above, we see that:

$$\text{Let: } y_1[n] = Ax_1[n] + B \text{ and } y_2[n] = Ax_2[n] + B$$

Additivity Test:

$$y[n] = y_1[n] + y_2[n]$$

$$y[n] = x_1[n^2] + x_2[n^2]$$

Given that both of the equations are the same, we can conclude that this function is linear.

2. The following two signals are available. Find the cross correlation.

$$x[n] = \{\dots, 0, 0, 2, -1, 3, 7, 1, 2, -3, 0, 0, \dots\}$$

$$y[n] = \{\dots, 0, 0, 1, -1, 2, -2, 4, 1, -2, 5, 0, 0, \dots\}$$

Answer:

Cross correlation is given by

$$r_{xy} = \sum_{n=-\infty}^{\infty} x[n]y[n-l] \forall l$$

As such, for the values of l as 0, 1 and -1:

$$r_{xy}(0) = 2 + 1 + 6 - 14 + 4 + 2 + 6 = 7$$

$$r_{xy}(1) = -1 - 3 + 14 - 2 + 8 - 3 = 13$$

$$r_{xy}(-1) = -2 - 2 - 6 + 28 + 1 - 4 - 14 = 0$$

You may follow the same method for the remainder of the values of l .

3. The fibonnaci sequence is defined by:

$$f[n] = f[n-1] + f[n-2]$$

(a) Develop a direct formula to calculate $f[n]$ only based on n **Answer:**

Let $f[n] = \alpha r^n$

$$\therefore \alpha r^n - \alpha r^{n-1} - \alpha r^{n-2} = 0$$

$$\frac{1 \pm \sqrt{5}}{2}$$

$$\therefore f[n] = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1 - \sqrt{5}}{2} \right)^n$$

(b) Show that $y[n]$ can also be represented by the LTI system $y[n] = y[n-1] + y[n-2] + x[n-1]$.

Answer :

When $x[n]$ is an impulse ($\delta(n)$), the sequence begins with 0 at origin, 1 at $n = 1$ and then continues as the fibonnaci sequence must go by. As such, this system is a valid representation of the Fibonnaci series.

4. Let $x[n]$ and $y[n]$ be two independent zero mean WSS random signals with autocorrelations R_{xx} and R_{yy} . Consider a random signal $v[n]$ obtained by a linear combination of x and y . Express that autocorrelation and cross correlations R_{vx} , R_{vy} and R_{vv} in terms of R_{xx} and R_{yy} .

Answer:

Let:

$$v[n] = ax[n] + by[n]$$

From here, we can easily find the autocorrelation:

$$R_{vv} = E(v[n+m]v[n]) = a^2R_{xx} + b^2R_{yy}$$